Activity No. 13	
Working with Files and Directories in Linux	
Course Code: CPE 201A	**Program:** Computer Engineering
Course Title: COMPUTER SYSTEM ADMINISTRATION AND TROUBLESHOOTING	**Date Performed:** October 20, 2026
Section: CPE11S2	**Date Submitted:** October 20, 2026
Name: Blizhel Torralba	**Instructor:** Sir Noel Jason Lusung

1. Objective/s:

This activity aims to navigate and manage files and directories using command line interface.

2. Intended Learning Outcome/s:

The students should be able to:

2.1 Demonstrate how to navigate home and system directories
2.2 Demonstrate how to how to list files and directories.
2.3 Demonstrate how to use globbing to manipulate files and directories.
2.4 Demonstrate how to create, move and delete files and directories.

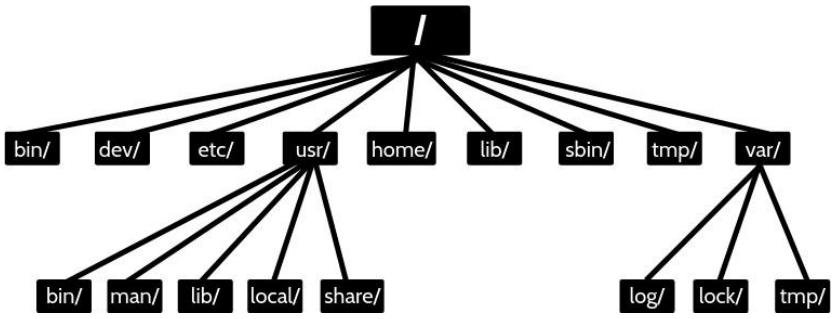
3. Discussion:

Files and Directories

The Linux filesystem is similar to other operating system's filesystems in that it contains *files* and *directories*. Files contain data such as human-readable text, executable programs, or binary data that is used by the computer.
Directories are used to create organization within the filesystem. Directories can contain files and other directories.

Directory Structure

On a Windows system, the *top level* of the directory structure is called ***My Computer.*** The Linux directory structure, called a ***filesystem***, also has a top level called the ***root directory*** (symbolized by the **slash /character**).



Home Directory

On most Linux distributions there is a directory called **home** under the root /directory. Under this **/home** directory there is a directory for each user on the system. When a user opens a shell, they should automatically be placed in their home directory. The user has the full control to create and delete additional files and directories in their home directory. Most other directories in a Linux filesystem are

protected with *file permissions*. The home directory has a special symbol used to represent it, the **tilde ~ character**.

The directory name is the same as the name of the user. So, a user named sysadmin would have a *home directory* called /home/sysadmin:

File and Directory Names

File and directory names in Linux can contain lower case and upper case letters, numbers, spaces and special characters. However, since many special characters have a special meaning in the Linux shell, it is good practice to not use spaces or special characters when naming files or directories. Spaces, for example, need the *escape character* \ to be entered correctly

Paths

A path is a list of directories separated by the / character. There are two types of paths: *absolute* and *relative*. For example, /home/sysadmin is a path to the home directory

A. Absolute Paths

Absolute paths allow the user to specify the exact location of a directory. Absolute paths always starts at the root directory, and therefore it always begins with the / character.

B. Relative Paths

A relative path gives directions to a file relative to the current location in the filesystem. The user must currently be in a directory that contains objects in the path. Relative paths start with the name of a directory.

Special Relative Paths

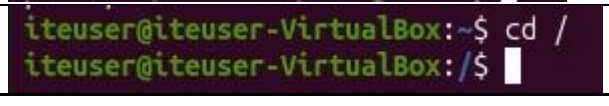
The single period . character always represents the current directory.
Two period .. characters always represents one directory higher relative to the current directory, sometimes referred to as the parent directory.

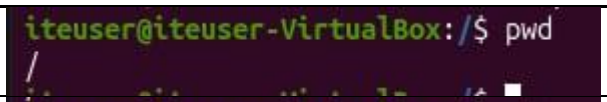
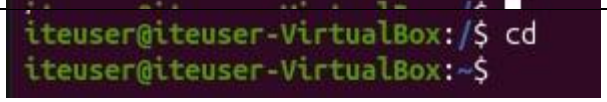
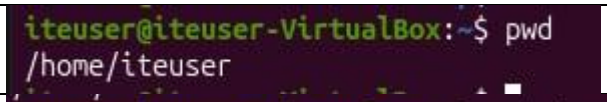
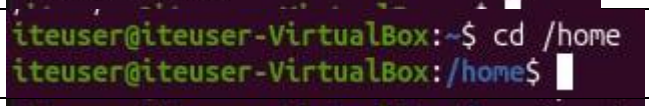
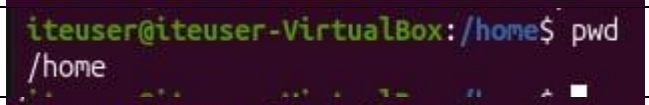
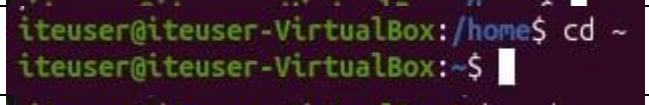
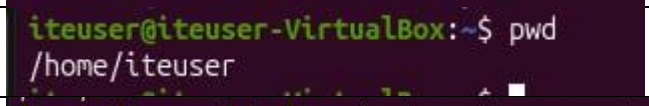
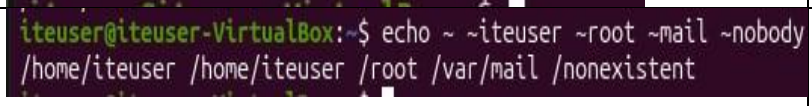
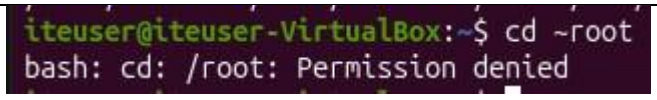
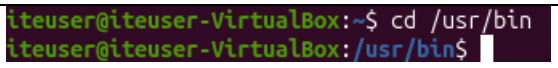
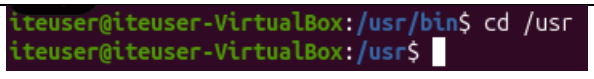
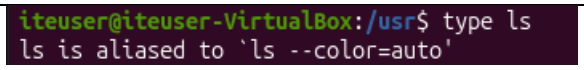
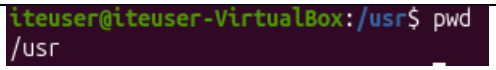
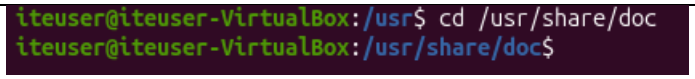
4. Resources:

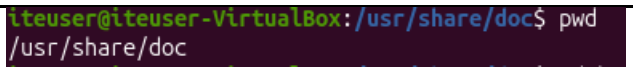
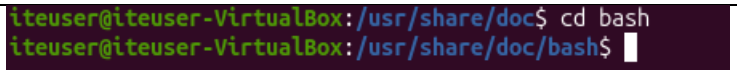
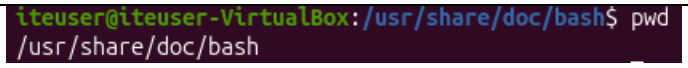
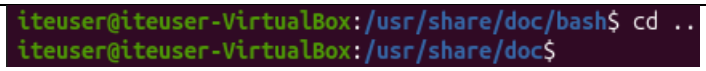
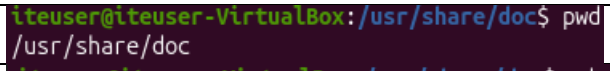
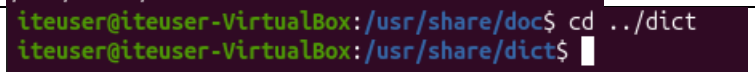
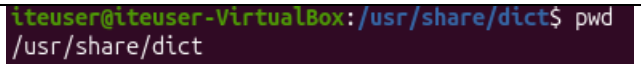
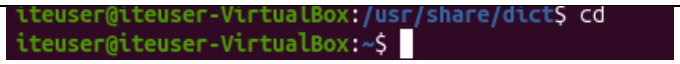
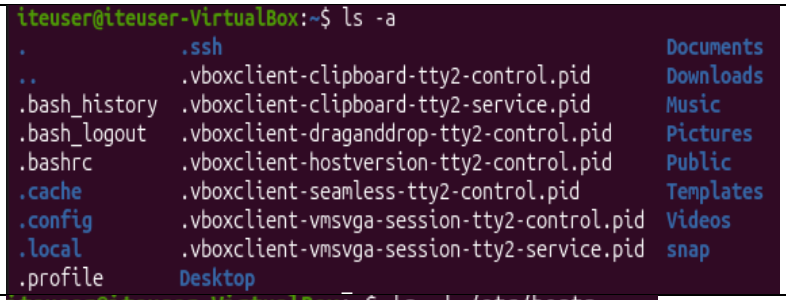
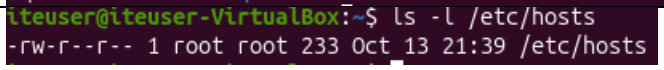
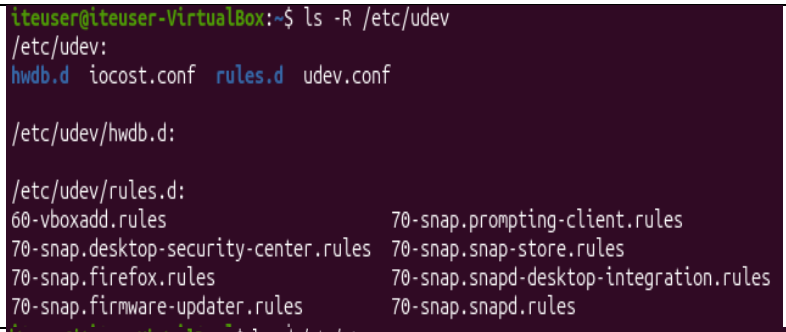
Personal Computer with installed Virtual Box
Ubuntu Server or Desktop virtual machine

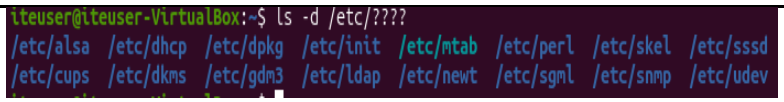
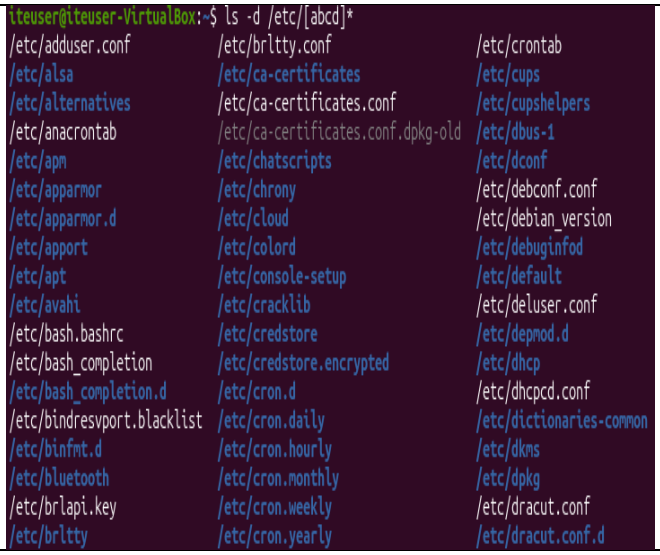
5. Procedure:

1. Login using your username and password.
2. Use terminal emulator application (if you are using desktop version)
3. Execute the following commands to navigate files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

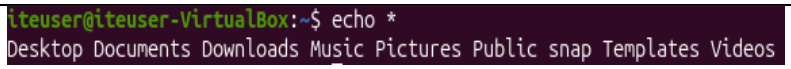
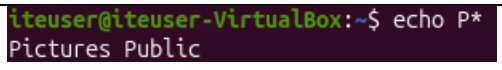
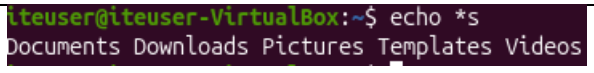
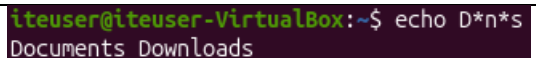
Command	Screenshot	Explanation
1. pwd		Print which folder you are currently in
2. cd /		Changes directory to (/)

3. pwd		Prints current directory
4. cd		Changes prompt/directory to (~)
5. pwd		Prints current directory
6. cd /home		Moves to /home directory
7. pwd		Prints current directory
8. cd ~		Goes to home directory
9. pwd		Prints current directory
10. echo ~ ~sysadmin ~root ~mail ~nobody where sysadmin is your username		Prints what each tilde shortcut expands to
11. cd ~root		Tries to go to root's home directory
12. cd /usr/bin		Moves to /usr/bin directory
13. pwd		Prints current directory
14. cd /usr		Moves to /usr directory
15. type ls		Shows how it interprets
16. pwd		Prints current directory
17. cd /usr/share/doc		Moves to folder where installed software keeps its documentations

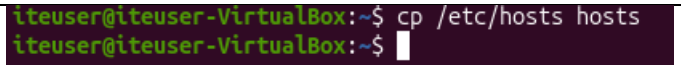
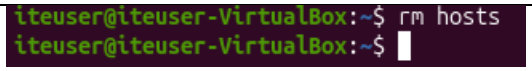
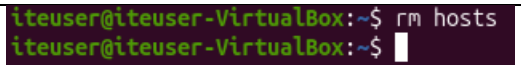
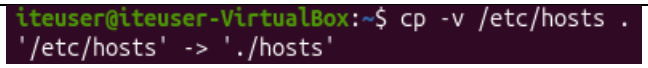
18. pwd		Prints current directory
19. cd bash		Moves to bash subfolder
20. pwd		Prints current directory
21. cd ..		Goes up one level
22. pwd		Prints directory
23. cd ../dict		Goes up one level and moves to dict
24. pwd		Prints current directory
25. cd		Returns you to home directory
26. ls		Lists files and folders in the current directory
27. ls -a		Lists all files including hidden files that starts with a dot
28. ls -l /etc/hosts		Shows long lists of hosts
29. ls -R /etc/udev		List all in /etc/udev and iits subdirectories
30. ls -d /etc/s*		Lists items in /etc that starts with s

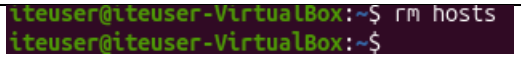
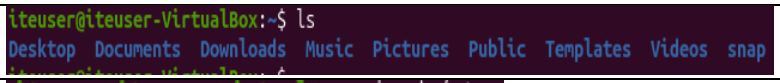
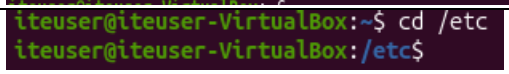
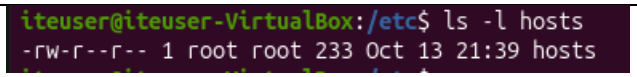
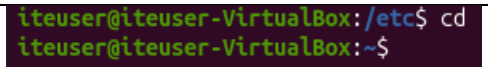
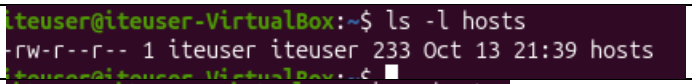
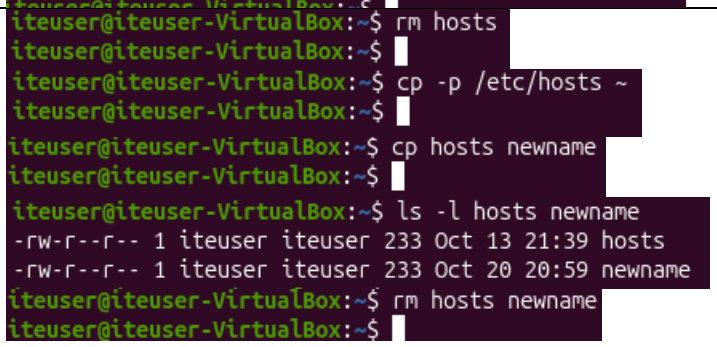
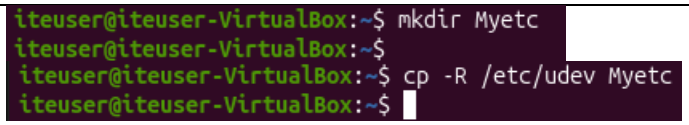
31. ls -d /etc/????		Lists items in /etc whose names are exactly 4 characters long
32. ls –d /etc/[abcd]*		Lists items in /etc whose names starts in a, b, c, and d

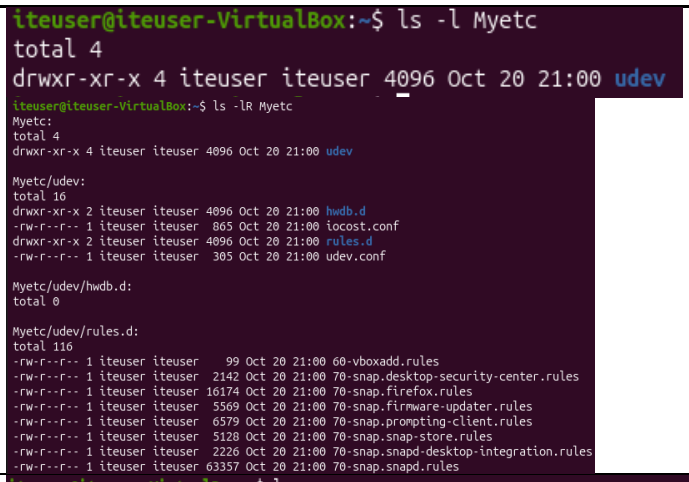
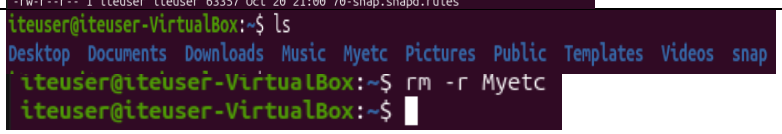
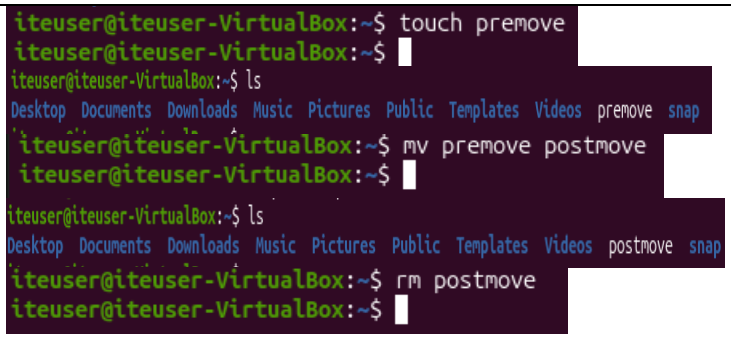
4. Execute the following commands to manage files and directories. Copy a screenshot as output after you execute the given command. Create a brief explanation of the command.

Command	Screenshot	Explanation
1. echo *		Prints names of all non-hidden files in the current directory
2. echo D*		Prints names that starts with D
3. echo P*		Prints names that starts with P
4. echo *s		Prints names that ends with s
5. echo D*n*s		Prints names

		that starts with D, contains n, and ends with s
6. echo ??????	iteuser@iteuser-VirtualBox:~$ echo ?????? Public Videos	Prints names that are exactly 6 characters long
7. echo D????????	iteuser@iteuser-VirtualBox:~$ echo D???????? Documents Downloads	Prints characters that starts with D and has 8 more characters
8. echo ?????*s	iteuser@iteuser-VirtualBox:~$ echo ?????*s Documents Downloads Pictures Templates Videos	Prints names with atleast 5 characters before it ends with s
9. echo [DP]*	iteuser@iteuser-VirtualBox:~$ echo [DP]* Desktop Documents Downloads Pictures Public	Prints names that starts with D or P
10. echo [!DP]*	iteuser@iteuser-VirtualBox:~$ echo [!DP]* Music snap Templates Videos	Prints names that do not start with D and P
11. echo [D-P]*	iteuser@iteuser-VirtualBox:~$ echo [D-P]* Desktop Documents Downloads Music Pictures Public	Prints names that starts with any letter from D to P
12. echo [!D-P]*	iteuser@iteuser-VirtualBox:~$ echo [!D-P]* snap Templates Videos	Prints names that do not start with letters

		from D to P
13. ls		Lists all files and folders in current directory
14. cp /etc/hosts hosts		Copies /etc/hosts in current directory as file named hosts
15. ls		Lists the directory and shows hosts
16. rm hosts		Removes hosts
17. ls		Lists the directory and shows that hosts is gone
18. cp –v /etc/hosts hosts		Copies file again
19. ls		Lists directory and shows hosts again
20. rm hosts		Removes hosts again
21. ls		Shows directory and confirms hosts is deleted
22. cp –v /etc/hosts .		Copies /etc/hosts into the directory

23. ls		Shows directory files and folders
24. rm hosts		Removes hosts again
25. ls		Confirms it is deleted
26. cd /etc		Moves to /etc directory
27. ls -l hosts		Shows list of hosts
28. cp –p hosts/home/ sysadmin		Copies hosts to home directory
29. cd		Returns to home directory
30. ls –l hosts		Shows list of hosts
31. rm hosts 32. cp -p /etc/hosts ~ 33. cp hosts newname 34. ls –l hosts newname 35. rm hosts newname		31. removes hosts 32. copies /etc/hosts to home directory 33. makes a copy of hosts names newname 34. shows list of both files 35. deletes both files
36. mkdir Myetc 37. cp –R /etc/udev Myetc 38. ls –l Myetc 39. ls –lR Myetc		36. creates new directory names Myetc

		37. copies /etc/udev directory into Myetc 38. shows Myetc 39. lists Myetc and all subfolders
40. ls 41. rm -r Myetc42.ls		40. lists currect directory 41. deletes Myetc
43. touch premove 44. ls 45. mv premove postmove 46. ls 47. rm postmove		43. Creates empty file named premove 44. shows premove 45. renames premove to postmove 46. shows postmove 47. deletes postmove

6. Supplementary Activity:

Copy screen shot(s) of the following tasks:

1. Create a directory CPE231 in the Documents directory.
2. Create a directory <lastname> and <firstname> under CPE231 directory.
3. Create two text files: (1) A.txt and (2) B.txt in the <lastname> directory.
4. Copy the content of <lastname> directory to <firstname> directory.
5. Delete the content of <lastname> directory.
6. Create a directory CPE231_backup in the Documents directory.
7. Copy the content of CPE231 to CPE231_backup directory.
8. Remove the CPE231 directory.

```
iteuser@iteuser-VirtualBox:~/Documents$ mkdir ~/Documents/CPE231
iteuser@iteuser-VirtualBox:~/Documents$ cd ~/Documents/CPE231
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ mkdir torralba blizhel
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ touch torralba/A.txt torralba/B.txt
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ cp torralba/* blizhel
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ rm torralba/*
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ mkdir ~/Documents/CPE231_backup
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ cp ~R ~/Documents/CPE231/* ~/Documents/CPE231_backup
cp: cannot stat '~R': No such file or directory
cp: -r not specified; omitting directory '/home/iteuser/Documents/CPE231/blizhel'
cp: -r not specified; omitting directory '/home/iteuser/Documents/CPE231/torralba'
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ ls ~R ~/Documents/CPE231_backup
ls: cannot access '~R': No such file or directory
/home/iteuser/Documents/CPE231_backup:
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ cp -R ~/Documents/CPE231/* ~/Documents/CPE231_backup
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ ls -R ~/Documents/CPE231_backup
/home/iteuser/Documents/CPE231_backup:
blizhel  torralba

/home/iteuser/Documents/CPE231_backup/blizhel:
A.txt  B.txt

/home/iteuser/Documents/CPE231_backup/torralba:
iteuser@iteuser-VirtualBox:~/Documents/CPE231$ rm -r ~/Documents/CPE231
iteuser@iteuser-VirtualBox:~/Documents/CPE231$
```

7. Conclusion:

In this activity, I learned how to navigate and manage files. I learned how to look for files, how to move directories, and create new files.

8. Assessment (Rubric for Laboratory Performance):

TIP Rubric E (1) (1) (1)

Criteria	Ratings		Pts
Performance Indicators 1. Apply appropriate techniques, skills, and modern tools to perform a discipline-specific engineering task.	**4 pts** **Very Satisfactory** Applies the most appropriate modern technique in performing discipline-specific engineering task exceeding the requirements.	**0 pts** **No Marks**	4 pts
Performance Indicators 2. Demonstrate skills in applying different techniques and modern tools to solve engineering problems.1. Apply appropriate techniques, skills, and modern tools to perform a discipline-specific engineering task.	**4 pts** **Very Satisfactory** Applies the most appropriate modern technique in performing discipline-specific engineering task exceeding the requirements.	**0 pts** **No Marks**	4 pts
Performance Indicators 3. Recognize the benefits and constraints of modern engineering tools.Demonstrate skills in applying different techniques and modern tools to solve engineering problems.1. Apply appropriate techniques, skills, and modern tools to perform a discipline-specific engineering task.	**4 pts** **Very Satisfactory** Applies the most appropriate modern technique in performing discipline-specific engineering task exceeding the requirements.	**0 pts** **No Marks**	4 pts
			Total Points: 12